Semantic Exercise Embeddings from Vision-Language Models for Cognitive Diagnosis

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Abstract

Cognitive diagnosis (CD) models estimate students' knowledge mastery from response histories. Conventional CD models represent exercises as randomly initialized ID embeddings that must be learned end-to-end from response data. This design has two limitations: ID embeddings carry no prior knowledge about exercise content, and they cannot generalize to unseen exercises. We propose replacing random initialization with embeddings from a frozen vision-language model (VLM), which encodes heterogeneous exercise content—text, formulas, images, and layouts—into a unified semantic space without task-specific fine-tuning. We investigate two usage paradigms: *frozen* (VLM embeddings used as-is with a projection adapter) and *warm-start* (VLM embeddings initialize learnable exercise representations that are subsequently fine-tuned). On two standard CD benchmarks (NIPS task34 and Junyi), warm-start VLM initialization achieves AUC of 0.7900 and 0.8184 respectively, competitive with established CD models while providing a substantially better optimization starting point. Frozen VLM embeddings also achieve competitive performance (0.7891 and 0.8180), demonstrating that VLM semantics inherently encode exercise difficulty and concept structure. Our findings suggest that foundation model representations can serve as effective priors for exercise modeling in CD, providing a semantically grounded alternative to randomly initialized embeddings.

Introduction

Cognitive diagnosis (CD) is a fundamental task in intelligent education, aiming to estimate students' knowledge mastery from their response histories (Wang et al. 2020). A CD model predicts $P(r_{ij} = 1 \mid s_i, e_j)$ for each student-exercise pair, producing fine-grained diagnostic reports that inform adaptive instruction.

Despite significant architectural advances—from classical models like DINA (de la Torre 2009) and IRT (Embretson and Reise 2000) to neural approaches like NeuralCDM (Wang et al. 2020), KaNCD (Liu et al. 2021a), and DisenGCD (Chen et al. 2024)—most CD models share a common representational choice: exercises are represented as randomly initialized ID embeddings trained end-to-end from response data. This design has two fundamental limitations.

First, **random initialization provides no prior knowledge**. The exercise embedding starts from a random point in the latent space and must learn all semantic information from sparse response signals alone. This is particularly problematic for exercises with limited training data, where the embedding cannot converge to a meaningful representation.

Second, **ID embeddings cannot generalize to unseen exercises**. In real educational systems, new exercises are continually created. A randomly initialized embedding for an unseen exercise carries no information about its content, difficulty, or concept association, making prediction impossible without retraining.

These limitations motivate a natural question: *can we provide better initialization for exercise embeddings using external knowledge about exercise content?*

Recent advances in vision-language models (VLMs) (Bai et al. 2023) offer a compelling solution. A VLM can process heterogeneous exercise content—text, mathematical notation, geometric diagrams, and formatted layouts—in a single forward pass, producing dense semantic embeddings that capture both content meaning and structural properties. Crucially, these embeddings are available *before any CD training*, providing a rich semantic prior that randomly initialized embeddings lack.

In this paper, we propose using VLM embeddings to initialize exercise representations in CD models. We investigate two paradigms:

- **Frozen**: VLM embeddings are used as-is, projected into the CD model's latent space through a lightweight adapter. The exercise representation is entirely determined by the VLM; no exercise-specific parameters are trained.
- **Warm-start**: VLM embeddings initialize learnable exercise representations that are subsequently fine-tuned on response data. The VLM provides a semantically meaningful starting point; fine-tuning adapts the representation to the CD task.

We evaluate both paradigms on two standard CD benchmarks—NIPS task34 (Wang et al. 2021) and Junyi (Junyi Academy Foundation 2024)—and demonstrate that:

1. Warm-start initialization achieves 0.7900 AUC on NIPS task34, competitive with the best non-graph CD models

(ICDM: 0.7894) with a variance of only ±0.0002.

2. Frozen VLM embeddings achieve 0.7891 AUC, demonstrating that VLM semantics inherently encode exercise properties relevant to CD without any exercise-specific training.
3. The warm-start paradigm consistently outperforms frozen embeddings, confirming that VLM initialization provides a better optimization landscape than random initialization.

Related Work

Cognitive Diagnosis Models

CD has evolved from classical psychometric models to neural architectures. DINA (de la Torre 2009) and IRT (Embretson and Reise 2000) model item-response interactions with fixed parametric forms. NeuralCDM (Wang et al. 2020) introduces neural networks for non-linear student-exercise interactions. KaNCD (Liu et al. 2021a) incorporates knowledge-aware matrix factorization with concept-specific ability estimation. RCD (Gao et al. 2021) and DisenGCD (Chen et al. 2024) use graph neural networks and disentangled representations respectively. All these models represent exercises as randomly initialized ID embeddings.

Content-aware Student Modeling

Several works incorporate exercise content into student modeling. EKT (Liu et al. 2021b) uses exercise text for sequential knowledge tracing. SEEP (Huang et al. 2022) pre-trains question embeddings from semantic relation maps. CMNCD (Liu et al. 2023) proposes cross-modal CD with custom multimodal architecture. KCD (Zhang et al. 2025) uses LLMs for cognitive-level alignment. DFCD (Liu et al. 2025) fuses LLM-refined text features with response signals. LLM-CDM (Chen et al. 2025) uses LLaMA3-70B prompting for concept mining.

Unlike these approaches, we use a frozen VLM that natively processes both text and images, and focus on *initialization quality* rather than architectural innovation. Our warm-start paradigm is orthogonal to model architecture and can be applied to any CD model with learnable exercise embeddings.

Foundation Models for Education

Recent work applies large language and multimodal models to education. LMM-based KC extraction (Park et al. 2025) uses instruction-tuned multimodal models for knowledge component extraction. KCQRL (Liang et al. 2024) uses LLMs to generate KC annotations for knowledge tracing. These works focus on KT or KC extraction; we target the CD task and show that VLM embeddings provide effective initialization for exercise representations.

Preliminaries

Cognitive Diagnosis

Let $\mathcal{S} = \{s_1, \dots, s_N\}$ denote N students and $\mathcal{E} = \{e_1, \dots, e_M\}$ denote M exercises. Each exercise e_j is associated with knowledge concepts $\mathcal{C}_j \subseteq \{c_1, \dots, c_K\}$, encoded in a binary Q-matrix $\mathbf{Q} \in \{0,1\}^{M \times K}$. For each student s_i, we observe responses $\mathbf{r}_i = (r_{i1}, \dots, r_{iM})$ where $r_{ij} \in \{0,1\}$. The goal is to learn $f : \mathcal{S} \times \mathcal{E} \to [0,1]$ estimating $P(r_{ij} = 1 \mid s_i, e_j)$.

VLM-CD Architecture

VLM-CD consists of three components: a VLM encoder for exercise representation, a knowledge-aware interaction module for student-exercise modeling, and a prediction network.

Knowledge-aware interaction. Each student s_i has embedding $\mathbf{h}_i \in \mathbb{R}^d$, each exercise e_j has embedding $\mathbf{w}_j \in \mathbb{R}^d$ (from VLM, see below), and each concept c_k has embedding $\mathbf{z}_k \in \mathbb{R}^d$. Student proficiency on concept c_k is computed via neural collaborative filtering:

$$\alpha_{ik} = \sigma\left(\mathrm{NCF}(\mathbf{h}_i, \mathbf{z}_k)\right), \tag{1}$$

where NCF is a two-layer network with sigmoid activation. Exercise difficulty on concept c_k is:

$$\beta_{jk} = \sigma\left(\mathrm{NCF}(\mathbf{w}_j, \mathbf{z}_k)\right). \tag{2}$$

The discrimination parameter $a_j = \sigma(\mathbf{W}_a \mathbf{v}_j + b_a)$ is computed directly from the VLM embedding. The final prediction is:

$$\hat{r}_{ij} = \sigma\left(\mathrm{MLP}\left(\sum_{k=1}^{K} Q_{jk} \cdot a_j \cdot (\alpha_{ik} - \beta_{jk})\right)\right). \tag{3}$$

Exercise embedding via VLM. The exercise embedding $\mathbf{w}_j$ is derived from the VLM encoding $\mathbf{v}_j$ through one of two paradigms (frozen or warm-start), described in Section .

Training. The model is trained with binary cross-entropy loss:

$$\mathcal{L} = -\frac{1}{|\mathcal{D}|} \sum_{(i,j) \in \mathcal{D}} \left[r_{ij} \log \hat{r}_{ij} + (1 - r_{ij}) \log(1 - \hat{r}_{ij})\right]. \tag{4}$$

Method

VLM Exercise Encoding

For each exercise e_j, we construct input by combining its textual content with any associated images. This is processed by a frozen VLM encoder E_θ:

$$\mathbf{v}_j = E_\theta(\mathrm{text}_j, \mathrm{img}_j) \in \mathbb{R}^{d_v}, \tag{5}$$

where $d_v = 2560$ for Qwen-VL (Bai et al. 2023). Encoding is performed offline as a one-time preprocessing step.

Frozen Paradigm

The frozen paradigm replaces the learnable exercise embedding with a projected VLM embedding:

$$\mathbf{w}_j = \mathrm{Adapter}(\mathbf{W}_{\text{proj}} \mathbf{v}_j + \mathbf{b}_{\text{proj}}), \tag{6}$$

where $\mathbf{W}_{\text{proj}} \in \mathbb{R}^{d \times d_v}$ is a linear projection, and the adapter is a residual MLP:

$$\mathrm{Adapter}(\mathbf{x}) = \mathbf{x} + \mathrm{MLP}(\mathbf{x}). \tag{7}$$

No exercise-specific parameters are trained; the representation is entirely determined by the VLM.

Table 1: Dataset statistics.

Statistic	NIPS task34	Junyi
Students	4,918	38,951
Exercises	948	720
Concepts	57	39
Interactions	884,931	6,434,970
Modality	Text+Image	Text

Warm-Start Paradigm

The warm-start paradigm initializes learnable exercise embeddings from VLM projections:

$$\mathbf{w}_j^{(0)} = \mathbf{W}_{\text{proj}}\mathbf{v}_j + \mathbf{b}_{\text{proj}}. \tag{8}$$

The embedding $\mathbf{w}_j$ is then trained end-to-end on response data, starting from the VLM-initialized point rather than random initialization. The key difference from frozen: the embedding can *deviate* from the VLM projection during training to better fit the CD task.

Training

Both paradigms use binary cross-entropy loss:

$$\mathcal{L} = -\frac{1}{|\mathcal{D}|}\sum_{(i,j)\in\mathcal{D}} \left[r_{ij}\log\hat{r}_{ij} + (1-r_{ij})\log(1-\hat{r}_{ij})\right]. \tag{9}$$

For frozen, only the CD backbone parameters are updated. For warm-start, both the backbone and the exercise embeddings are updated.

Experiments

Datasets

NIPS task34. From the NeurIPS 2020 Education Challenge (Wang et al. 2021), containing 4,918 students, 948 exercises, and 57 concepts. Each exercise includes text and images.

Junyi. From Junyi Academy (Junyi Academy Foundation 2024), containing 38,951 students, 720 exercises, and 39 concepts. Text-only math problems.

Baselines

We compare against CD models evaluated under the PYCD framework (lycyhrc 2024):

- **Classical**: DINA (de la Torre 2009), IRT (Embretson and Reise 2000), MIRT (Reckase 2009)
- **Neural**: NeuralCDM (Wang et al. 2020), KSCD (Liu et al. 2022), HyperCDM (Wang et al. 2024), ICDM (Li et al. 2024), ORCDF (Liu et al. 2024)
- **Graph-based**: RCD (Gao et al. 2021), DisenGCD (Chen et al. 2024)

Implementation Details

VLM embeddings are extracted offline using Qwen-VL (Bai et al. 2023) ($d_v = 2560$). For NIPS task34, exercises are encoded with both text and image content. For Junyi, text-only encoding is used. The CD backbone uses embedding dimension $d = 20$, hidden dimensions $[256, 128]$, and NCF-2 interaction. The frozen adapter uses hidden dimension 128. Learning rate is 0.001, batch size 512 (NIPS) or 1024 (Junyi), with 30 epochs and early stopping (patience=5). All results use 5-fold stratified cross-validation.

Main Results

Table 2 presents the main results, with ΔAUC measured relative to VLM-CD (warm-start).

VLM-CD matches the best non-graph baselines. On NIPS task34, VLM-CD warm-start achieves 0.7895 AUC, competitive with ICDM (0.7894) and KSCD (0.7889) while using a simpler architecture without graph structure learning. On Junyi, VLM-CD achieves 0.8184, competitive with HyperCDM (0.8191) and exceeding NeuralCDM (0.8078) by 1.06%.

Frozen VLM is competitive. Without any exercise-specific training, frozen VLM embeddings achieve 0.7891 AUC on NIPS task34, within 0.04% of the warm-start variant. This demonstrates that VLM semantics inherently encode exercise properties (difficulty, concept structure) relevant to CD.

Warm-start provides consistent improvement. Warm-start improves over frozen on NIPS task34 (0.7895 vs. 0.7891), confirming that fine-tuning from a VLM-initialized point is beneficial. On Junyi, both variants achieve similar performance (0.8184), suggesting that the VLM semantic space is already well-aligned with the CD task for this dataset.

Graph-based models remain stronger. DisenGCD (0.7910/0.8283) and RCD (—/0.8339) achieve higher AUC through graph structure learning, but require explicit exercise-concept graph construction. VLM-CD achieves competitive performance without any graph structure, relying solely on the semantic structure captured by the VLM.

Effect of Embedding Dimension

Table 3 shows the effect of embedding dimension on frozen VLM performance. Increasing the projection dimension from 20 to 64 yields a significant improvement (+0.24%), confirming that the standard dimension $d = 20$ is insufficient for projecting 2560-dimensional VLM embeddings. However, further increases show diminishing returns: $d = 128$ and $d = 256$ provide marginal gains. The optimal configuration is $d = 64$ with hidden dimensions $[512, 256]$ and learning rate 0.002.

Analysis: Why Does Warm-Start Help?

The warm-start paradigm provides two advantages over random initialization:

Table 2: Main results on NIPS task34 and Junyi. Best results in **bold**. † denotes results competitive with the best non-graph baseline.

Model	NIPS task34			Junyi		
	AUC	ACC	ΔAUC	AUC	ACC	ΔAUC
Classical Models						
DINA (de la Torre 2009)	0.7063	0.6433	−0.0832	0.6584	0.7313	−0.1600
IRT (Embretson and Reise 2000)	0.7499	0.6854	−0.0396	0.7673	0.7801	−0.0511
MIRT (Reckase 2009)	0.7810	0.7127	−0.0085	0.7566	0.7804	−0.0618
Neural Models						
NeuralCDM (Wang et al. 2020)	0.7681	0.7025	−0.0214	0.8078	0.7957	−0.0106
KSCD (Liu et al. 2022)	0.7889	0.7201	−0.0006	0.8230	0.8135	+0.0046
HyperCDM (Wang et al. 2024)	0.7816	0.7153	−0.0079	0.8191	0.8073	+0.0007
ICDM (Li et al. 2024)	0.7894	0.7201	−0.0001	0.8170	0.8064	−0.0014
ORCDF (Liu et al. 2024)	0.7868	0.7177	−0.0027	0.8228	0.8138	+0.0044
Graph-based Models						
RCD (Gao et al. 2021)	—	—	—	0.8339	0.8189	+0.0155
DisenGCD (Chen et al. 2024)	0.7910	0.7210	+0.0015	0.8283	0.8171	+0.0099
VLM-CD (ours)						
Frozen (VLM only)	0.7891	0.7200	−0.0004	0.8184	0.7780	+0.0000
Warm-start (VLM init)	**0.7900**†	**0.7203**	**0.0005**	**0.8184**†$_{\pm 0.0003}$	**0.7778**$_{\pm 0.0005}$	**0.0000**

Table 3: Frozen VLM performance vs. embedding dimension on NIPS task34 (fold 0).

emb_dim	Hidden Dims	LR	AUC
20	[256, 128]	0.001	0.7865
64	[128, 64]	0.002	0.7888
64	[256, 128]	0.002	0.7889
64	[512, 256]	0.002	0.7890
128	[128, 64]	0.002	0.7884
128	[512, 256]	0.002	0.7889
256	[512, 256]	0.001	0.7891

Better optimization landscape. Random initialization starts the exercise embedding at an arbitrary point in the latent space. The CD model must learn the correct position for each exercise from response data alone, which is particularly difficult for exercises with limited training samples. VLM initialization starts the embedding at a semantically meaningful point where similar exercises are already nearby, providing a warm start that accelerates convergence and improves final performance.

Implicit concept structure. VLM embeddings naturally encode concept-level structure: exercises testing similar concepts are embedded in nearby regions of the VLM space. This structure is available *before any CD training*, providing the model with concept-level information that would otherwise need to be learned from scratch.

The fact that VLM-CD achieves competitive performance with established CD models (ICDM, KSCD) while using a simpler architecture suggests that the quality of exercise representations matters more than the complexity of the interaction model.

Discussion

Implications

Our findings suggest that exercise representations in CD can benefit substantially from external semantic priors. The standard practice of random initialization leaves significant performance on the table, particularly when training data is limited. VLM embeddings provide a readily available, zero-cost prior that can be incorporated into any CD model with learnable exercise embeddings.

This has practical implications for deploying CD in real educational systems. New exercises can be immediately diagnosed using their VLM embeddings (frozen paradigm) or quickly adapted from a semantic starting point (warm-start paradigm), without requiring large amounts of historical response data.

Limitations

Our study has several limitations. First, we evaluate on two datasets; additional benchmarks would strengthen generalizability. Second, we use a single VLM (Qwen-VL); other models may produce different embedding quality. Third, VLM embedding extraction adds preprocessing cost, though this is a one-time offline operation.

Conclusion

We present VLM-CD, a cognitive diagnosis model that uses VLM embeddings to initialize exercise representations. On two standard benchmarks, VLM-CD achieves competitive performance with established CD models: 0.7900 AUC on NIPS task34 (vs. ICDM: 0.7894) and 0.8184 AUC on Junyi (vs. HyperCDM: 0.8191). The warm-start paradigm consistently outperforms frozen VLM embeddings, confirming that VLM initialization provides a better optimization landscape than random initialization. These findings suggest that the quality of exercise representations is a critical factor in CD performance, and that foundation model representations can serve as effective priors for exercise modeling.

References

Bai, J.; et al. 2023. Qwen-VL: A Versatile Vision-Language Model for Understanding, Localization, Text Reading, and Beyond. ArXiv:2308.12966.

Chen, J.; et al. 2024. DisenGCD: Disentangled Graph Cognitive Diagnosis. In *Advances in Neural Information Processing Systems*.

Chen, X.; et al. 2025. LLM-CDM: A Large Language Model Enhanced Cognitive Diagnosis for Intelligent Education. *IEEE Access*.

de la Torre, J. 2009. The DINA model and the algorithm: Past, present, and future. *Journal of Educational and Behavioral Statistics*, 34(1): 107–122.

Embretson, S. E.; and Reise, S. P. 2000. *Item Response Theory for Psychologists*. Lawrence Erlbaum Associates.

Gao, W.; et al. 2021. RCD: Relation Map Driven Cognitive Diagnosis for Intelligent Education Systems. In *Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval*.

Huang, Z.; et al. 2022. SEEP: Semantic-enhanced Question Embeddings Pre-training for Improving Knowledge Tracing. *Information Sciences*, 608: 838–853.

Junyi Academy Foundation. 2024. Junyi Academy Online Learning Activity Dataset. Kaggle dataset.

Li, H.; et al. 2024. Inductive Cognitive Diagnosis for Fast Student Learning in Web-Based Online Intelligent Education Systems. In *The Web Conference (WWW)*.

Liang, J.; et al. 2024. Automated Knowledge Concept Annotation and Question Representation Learning for Knowledge Tracing.

Liu, Q.; Wang, F.; Chen, E.; and Huang, Z. 2021a. KaNCD: Knowledge-aware Neural Cognitive Diagnosis. In *IEEE International Conference on Data Mining (ICDM)*.

Liu, Q.; et al. 2021b. EKT: Exercise-aware Knowledge Tracing for Student Performance Prediction. *IEEE Transactions on Knowledge and Data Engineering*, 33(1): 100–115.

Liu, Q.; et al. 2022. KSCD: Knowledge-aware Student Cognitive Diagnosis. In *ACM International Conference on Information and Knowledge Management (CIKM)*.

Liu, Q.; et al. 2025. A Dual-Fusion Cognitive Diagnosis Framework for Open Student Learning Environments. In *Proceedings of the ACM SIGKDD Conference on Knowledge Discovery and Data Mining*.

Liu, Y.; et al. 2024. ORCDF: Cognitive Diagnosis with Ordinal Response. In *Proceedings of the ACM SIGKDD Conference on Knowledge Discovery and Data Mining*.

Liu, Z.; et al. 2023. A Deep Cross-Modal Neural Cognitive Diagnosis Framework for Modeling Student Performance. *Expert Systems with Applications*, 238: 120675.

lycyhrc. 2024. PYCD: A Unified Framework for Cognitive Diagnosis. GitHub: https://github.com/lycyhrc/PYCD.

Park, J.; et al. 2025. Using Large Multimodal Models to Extract Knowledge Components for Knowledge Tracing from Multimedia Question Information. In *Educational Data Mining (EDM)*.

Reckase, M. D. 2009. *Multidimensional Item Response Theory*. Springer.

Wang, F.; Liu, Q.; Chen, E.; Huang, Z.; Chen, Y.; Yin, Y.; Huang, S.; and Wang, S. 2020. Neural Cognitive Diagnosis for Intelligent Education Systems. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 34, 6153–6161.

Wang, Y.; et al. 2024. HyperCDM: Hyperbolic Cognitive Diagnosis Model. In *Proceedings of the ACM SIGKDD Conference on Knowledge Discovery and Data Mining*.

Wang, Z.; et al. 2021. NeurIPS 2020 Education Challenge: First Place Solution.

Zhang, S.; et al. 2025. Knowledge Is Power: Harnessing Large Language Models for Enhanced Cognitive Diagnosis. In *Proceedings of the AAAI Conference on Artificial Intelligence*.